

Continuous model calibration framework for smart-building digital twin: A generative model-based approach

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ABSTRACT

Smart building digital twins represent a significant paradigm shift to optimize building operations, thereby reducing their substantial energy consumption and emissions through digitalization. The objective is to virtually replicate existing buildings' static and dynamic aspects, leveraging data, information, and models spanning the entire life cycle. The virtual replica can then be employed for intelligent functions, including real-time monitoring, autonomous control, and proactive decision-making to optimize building operations. To enable proactive decisions, models within the digital twin must continually evolve with changes in the physical building, aligning their outputs with real-time measurements through calibration. This continuous updating requires real-time physical measurements of model inputs. However, challenges arise in the uncertain conditions of buildings marked by sensor absence, malfunctions, and inherent limitations in measuring certain variables. This study introduces a novel calibration framework for physics-based models, addressing the challenges of continuous model calibration in smart-building digital twins while considering the uncertain environment of physical buildings. Within this framework, a novel generative model-based architecture is proposed. This architecture enables a fast and scalable solution while quantifying uncertainty for reliable calibration. Furthermore, a continuous model calibration procedure is presented based on a pre-trained generative calibrator model. A comprehensive evaluation was conducted via a case study employing a building energy model and multiple experiments. The experimental results demonstrated that the proposed framework effectively addresses the challenges of continuous model calibration in smart-building digital twins. The calibrator model accurately quantified uncertainties in its predictions and solved a single calibration problem in an average time of 0.043 second. For facility-level electricity consumption, Coefficient of Variation Root Mean Squared Error (CVRMSE) values of 6.33%, 10.18%, and 10.97% were achieved under conditions of observations without noise or missing data, with noise, and with noise and missing data, respectively. Similarly, for facility-level gas consumption, the corresponding values were 18.75%, 20.53%, and 20.7%. The CVRMSE scores in both cases met the standard hourly thresholds for building energy model calibration.

1. Introduction

Buildings are significant contributors to global energy consumption and greenhouse gas emissions, accounting for approximately 40% of total energy usage and one-third of global emissions [1–3]. Consequently, integrating modern technologies such as the Internet of Things (IoT) and Artificial Intelligence (AI) in the built environment, particularly in buildings, has emerged as a potential solution to reduce energy consumption and emissions. Despite ongoing efforts, recent data from the International Energy Agency (IEA) indicates a continuing upward trend in energy consumption and emissions [4,5] emphasizing the need for more advanced, holistic technologies. In light of this need, smart-building digital twins have emerged, marking a significant shift toward

building digitalization, which has shown immense potential [6] in achieving global net-zero energy buildings and emission goals.

A digital twin is a continuously updated model that closely mirrors its physical counterpart through real-time data and can precisely predict its future states. The concept is built on three fundamental components: the physical entity, its virtual equivalent, and the bidirectional connectivity facilitating uninterrupted data flow between them [7–12]. Specifically, smart-building digital twins aim to virtually replicate the static and dynamic aspects of a smart building throughout its entire life cycle phases (i.e., design, construction, operation, and maintenance) [10,13–17] leveraging data, information

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and models. The smart-building digital twin can then enable intelligent functions such as real-time monitoring, autonomous control, and proactive decision-making to optimize buildings' operation [8,9,18,19].

Despite the advantage of smart-building digital twins, several challenges must be addressed before fully utilizing their potential. Such challenges stem from the inherent characteristics of buildings, such as their massive scale, heterogeneous environments, uniqueness, and long operational periods [20,21]. Unlike product digital twins created in the controlled environments of manufacturing, building digital twins must be developed on-site using data, information, and models from the whole life cycle [20]. One of the primary challenges in smart-building digital twins is updating models with the real-time state of their physical counterparts. This is crucial for ensuring that the models consistently mirror and accurately predict the evolving state of the physical building.

Models are fundamental elements in the digital twin's virtual layer, accurately representing various aspects of a physical building. Unlike most black-box models, physics-based models especially offer unmatched potential owing to their interpretability. A physics-based model refers to models derived from fundamental equations that govern energy, heat, and mass balance, such as building energy models [22]. These models enable high-fidelity simulations within the virtual layer of the smart-building digital twin, providing valuable insights for "what-if" scenarios in predicting the behavior of the physical building [23,24].

Physics-based modeling in smart-building digital twinning encounters challenges due to the unavailability of data for the verification of model outputs, such as electricity consumption in building energy models against physical measurements, i.e., whether the modeling environment is observed (physically measured) or unobserved (not measured in the physical building) [25]. In addition, the unobservability of certain influential inputs of the physics-based models requires continuous estimation to align the output of the model with physically measured data, a process referred to as "in-situ calibration" [20,21]. Physics-based models are considered calibrated when the discrepancy between their output and the physical measurements falls below a predefined threshold [20,26,27].

Existing calibration methods, particularly for physics-based building energy models, fall into three main categories: mathematical optimization search-based, surrogate-based (meta-model), and Bayesian inference approaches [26,28]. Optimization search-based approaches [26, 27,29–36] address the model calibration problem through an iterative process searching for unknown model inputs by employing methods such as gradient-decent, or evolutionary algorithms. The surrogate-based model calibration approaches [26,37–43] use meta-models to minimize the high computational costs associated with optimization search-based approaches by replacing the complex physics-based models with an approximate black-box model. On the other hand, Bayesian-based approaches [41,42,44–47] solve the calibration problem by integrating expert knowledge as priors, employing Bayes' theorem [48] to merge this with the likelihood of observed data, then refining initial assumptions into posterior samples [49].

Existing calibration methods for physics-based models have significant limitations in meeting the continuous calibration needs of smart-building digital twins. Iterative approaches, such as optimization search and Bayesian methods, are limited in the real-time updating of physics-based models due to their resource-intensive computation, slowness, and limited scalability for calibrating high-dimensional input–output models. Additionally, optimization search methods cannot quantify uncertainty in the estimated calibration solutions, making them unreliable for decision-making. Furthermore, existing calibration approaches did not account for the inherent challenges of sensing in building operation, characterized by random malfunctions and measurement noise.

Considering the aforementioned research gaps, this study proposes a calibration framework for physics-based models tailored to overcome the challenges of continuous model calibration in smart-building digital twins. The contributions (C) of this study are summarized as follows:

- C_1 : A novel probabilistic calibration framework for physics-based models, specifically devised to address the continuous and real-time model calibration requirements in smart-building digital twins.
- C_2 : A novel generative model architecture to enable continuous, fast, and more scalable calibration of physics-based models while quantifying the uncertainty of calibration solutions and taking into account uncertain sensing conditions.
- C_3 : A continuous model calibration procedure for physics-based models, based on a pre-trained calibrator model that utilizes ongoing physical sensor measurements and avoids computationally expensive search in the calibration process.

The remainder of this paper is organized as follows: Section 2 introduces key background concepts, followed by a review of related works in Section 3. Section 4 details the proposed model calibration framework, and Section 5 describes a comprehensive evaluation procedure through a case study, including results and a discussion of the main findings. Finally, Section 6 presents the conclusion and highlights directions for future works.

2. Theoretical background

This section outlines foundational concepts for the calibration framework discussed in Section 4. The background information on Normalizing Flows, affine coupling blocks, Conditional Invertible Neural Networks (cINN), and denoising autoencoders constitute the generative model architecture presented in Section 4.

2.1. Normalizing flows

Normalizing Flows (NF) are generative models that use invertible transformations to convert simple distributions, such as Gaussian, into more complex ones [50–53]. Given a random variable Z with base distribution $p_Z(z)$, and a transformation function $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $X = f(Z)$ and $Z = f^{-1}(X)$, the distribution of the more complex random variable X , denoted $p_X(x)$, can be determined using the change of variables theorem as shown in Eq. (1) [50,52,54,55]. In Eq. (1), f^{-1} is the inverse of the transformation function, and $\frac{\partial f^{-1}}{\partial x}$ is the Jacobian matrix, whose determinant quantifies the change in volume due to the transformation.

$$p_X(x) = p_Z(f^{-1}(x)) \left| \det \left(\frac{\partial f^{-1}}{\partial x} \right) \right| \quad (1)$$

2.2. Affine coupling blocks

The affine coupling blocks are popular normalizing flow architectures introduced by Dinh et al. in their work on real-valued non-volume preserving (RealNVP) transformations [51]. These transformations are notable for their efficient computation of the Jacobian determinant, exact log-likelihood evaluation, and sampling capabilities [51].

The forward process of the transformations in affine coupling blocks is summarized by Eq. (2). In each coupling block, the input i is split into two parts: i_1 and i_2 . The first coupling layer transforms i_1 through affine transformations, applying scale and translation factors where $\odot$ represents element-wise multiplication. These factors are derived by passing i_2 , the second half of the input through arbitrary neural networks, s_1 and t_1 , resulting in the transformed output O_1 . Subsequently, in the second coupling layer of the block, i_2 undergoes similar transformations. Here, the scale and shift factors s_2 and t_2 are calculated based on the previous output of the affine coupling layer. This process can be extended with multiple coupling blocks stacked sequentially for more complex transformations [51,56]. The inverse process is given by Eq. (3) where $\oslash$ represents element-wise division [56].

$$O_1 = i_1 \odot \exp(s_1(i_2)) + t_1(i_2) \quad (2)$$

$$\begin{aligned} O_2 &= i_2 \odot \exp(s_2(O_1)) + t_2(O_1) \\ i_2 &= (O_2 - t_2(O_1)) \oslash \exp(s_2(O_1)) \\ i_1 &= (O_1 - t_1(i_2)) \oslash \exp(s_1(i_2)) \end{aligned} \quad (3)$$

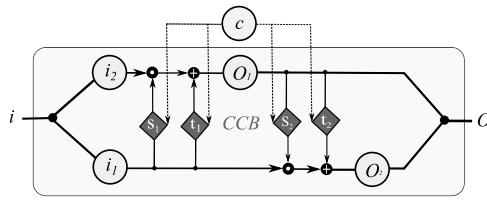


Fig. 1. Conditional coupling block (CCB).
Source: Adapted from [56].

2.3. Conditional Invertible Neural Networks (cINN)

The Conditional Invertible Neural Network (cINN) architecture introduces conditioning inputs into affine coupling blocks while preserving their bijective nature [56]. Fig. 1 shows the configuration of a single conditional coupling block, where the conditioning inputs, denoted by c , are concatenated with the inputs of the sub-networks.

Conditional Invertible Neural Networks (cINNs) are trained by minimizing the negative log-likelihood, thereby maximizing the likelihood of the data under the model [56]. The loss can be derived from Eq. (1), taking the logarithm of both sides, assuming a Gaussian prior on z , with a conditioning input c_i for a training sample i , and the network parameters θ , as shown in Eq. (4). The first term of this equation represents the log probability of the data under the transformation defined by the flow model where f is the transformation function, J_i is the Jacobian determinant at sample i , and θ denotes the network parameters. This term is averaged over N training samples. The second part of the loss introduces an L2 regularization term on the network weights, which helps control model complexity and prevent overfitting.

$$\mathcal{L}_{cINN} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{2} \|f(x_i; c_i, \theta)\|_2^2 - \log |J_i| \right) + \tau \|\theta\|_2^2 \quad (4)$$

2.4. Denoising autoencoders

Denoising autoencoders are a special type of autoencoder consisting of encoder, bottleneck, and decoder components. They are trained to learn robust representations by systematically corrupting the input data and trying to reconstruct it [57–59]. The corrupting noise to the input can be additive Gaussian or random masking.

Similar to vanilla autoencoders, denoising autoencoders can be trained using the Mean Squared Error (MSE) reconstruction loss, defined in Eq. (5), where X denotes the original input and $\hat{X}$ represents the reconstructed output. Here, X_i and $\hat{X}_i \in \mathbb{R}^D$ in a D -dimensional space, and the overall loss is the average over N training samples.

$$L_{MSE}(X, \hat{X}) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^D (X_{ij} - \hat{X}_{ij})^2 \quad (5)$$

3. Related work

Existing calibration methods for building-related models fall into three categories: mathematical optimization search-based, surrogate-based (meta-model), and Bayesian inference approaches [26,28].

3.1. Optimization search-based approaches

Optimization search-based approaches iteratively search calibration solutions to minimize the discrepancy between measured and simulated outputs. Many existing studies focus on calibrating building energy models, which are often over-parameterized and underdetermined. The most commonly used methods include gradient descent [26], genetic algorithms [27,29–32], and particle swarm optimization [33–36]. For instance, Herbringer et al. [26] implemented a gradient descent-based optimization using a multi-layer perceptron artificial neural network

surrogate. This approach calibrated a building energy model with hourly energy data by solving for fourteen unknown building parameters. Genetic Algorithms (GA) and particle swarm optimization are, on the other hand, gradient-free optimization capabilities, making them more effective than gradient-based optimization for building energy model calibration due to the need to minimize a cost function evaluated by external simulation programs [27].

The optimization-based methods have limitations in terms of high computational cost, as they require running multiple parameter sample combinations until convergence criteria are met. This challenge amplifies with an increase in the number of parameters to be calibrated. Furthermore, these deterministic approaches typically yield a single solution for the calibration problem and do not account for the uncertainty of estimated parameters. This is especially relevant in physics-based models, where equi-finality and uncertainty are prevalent, reducing confidence in the calibration solutions obtained.

3.2. Surrogate model-based approaches

The surrogate-based model calibration approaches aim to mitigate the high computational costs associated with optimization search-based approaches. This is achieved by utilizing computationally efficient surrogate models to emulate complex physics-based models, thereby avoiding running these intricate models throughout the optimization process [26,60]. Numerous studies have applied surrogate-based optimization in building energy model calibration [26,37–43]. While these surrogate-based approaches significantly enhance computational efficiency, the inherent iterative process of solving the calibration problem persists.

Furthermore, all limitations associated with optimization search-based approaches remain challenging, except executing complex physics models when surrogate models are employed in optimization-search-based calibration. Additionally, the accuracy of the surrogate models in accurately representing the complex physics-based models is a crucial factor in determining the accuracy of the calibration results.

3.3. Bayesian-based approaches

Bayesian-based calibration approaches have become increasingly popular in building energy model calibration, as evidenced by numerous studies [41,42,44–47] and virtual sensor model calibration in building digital twin context [21,61–67].

Despite the potential advantage of the Bayesian approach in enhancing confidence in calibration outcomes, it has significant drawbacks when used with computationally expensive statistical methods such as Markov Chain Monte Carlo (MCMC). This limitation is especially notable when multiple evaluations of complex building energy models are needed, making it less scalable [27]. Therefore, employing such an approach for continuous and fast model updates in smart-building digital twins becomes less efficient.

4. Proposed framework

The proposed framework for calibrating physics-based models in smart building digital twins employs probabilistic inverse models, specifically generative models. We refer to such models as calibrator models to specify their role in calibrating the physics-based model. The proposed Generative Model-based Calibration Framework for Physics-based Models in Smart-Building Digital Twins (**GenPhysiCal**) is shown in Fig. 2, and the subsequent subsections detail each component in the framework.

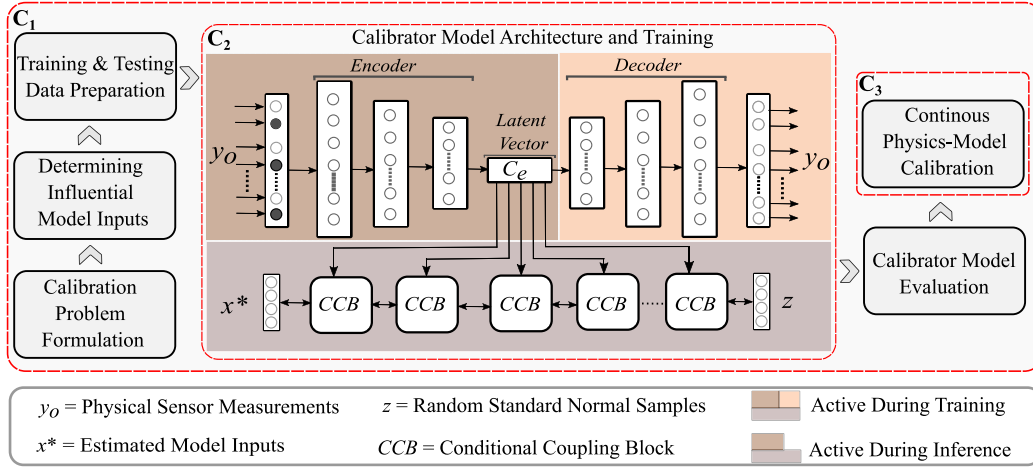


Fig. 2. Generative Model-based Calibration Framework for Physics-based Models in Smart-Building Digital Twins (GenPhysiCal).

4.1. Calibration problem formulation

Given a physics-based (white-box) model to be employed in the smart building digital twin, denoted as $M : x \rightarrow y$, which maps its model inputs $x \in \mathbb{R}^S$ into outputs $y \in \mathbb{R}^D$, according to a forward function $y = M(x)$ where S and D represent the dimensions of the inputs and outputs, respectively. Here, the model inputs x are a set of inputs with direct influence on the output of the physics-based model, i.e., a change in these model inputs results in a change in the output of the physics-based model and are not measured in the physical building. It is also assumed that the output variables of the physics-based model are observable in the physical building by sensors denoted by $y_o \in \mathbb{R}^D$. However, unlike the forward model outputs, the physical measurements y_o are subjected to noise expressed as $y_o = y + \epsilon$ where ϵ represents observation noise in the physical building.

Therefore, the first objective is solving for the posterior distribution denoted by $p(x | y_o)$ of the unobserved model inputs, which can be defined by the Bayesian formulation given in Eq. (6). where the forward model serves as the likelihood function denoted by $p(y_o | x)$, a prior distribution $p(x)$, the marginal likelihood of the physically-observable variables $p(y_o)$. The additional implicit goal in the calibration problem is estimating the values of the unobservable model inputs x that minimize the distance between the outputs of the physics model y and the physically measured variables y_o which can be expressed as Eq. (7) where L is some distance function that measures the discrepancy between the simulated and measured variables x^* is the value of unobserved model input that minimizes the distance metric.

$$P(x|y_o) = \frac{P(y_o|x)P(x)}{P(y_o)} \quad (6)$$

$$x^* = \underset{x}{\operatorname{argmin}} L(M(x), y_o) \quad (7)$$

4.2. Determining influential model inputs

As defined in the calibration problem formulation, the focus is on estimating model inputs that significantly affect its output. The GenPhysiCal framework particularly emphasizes dynamic inputs that vary during the operation of a building, in contrast to static parameters of the physics-based model. While these static parameters can also be influential, they remain constant once validated unless significant changes occur in the physical building due to aging or retrofits.

Additionally, the dynamic and influential model inputs are assumed not physically measured, often due to the absence of suitable sensors, malfunctioning equipment, or the complexity involved in measuring certain variables, such as infiltration rates and occupant numbers. As

a result, the influential model inputs are termed “unobservable model inputs”, distinguishing them from the static parameters.

In determining the influence of the unobservable model inputs, the output variables of interest from the physics model that are targeted to be calibrated guide the process. In this regard, domain knowledge about the sensitivity of the physics model outputs for changes in the model inputs can be leveraged, or a more formal approach is the use of global sensitivity analysis techniques to select the unobservable model inputs with a strong influence on the output of interest from the physics model. Global sensitivity analysis is particularly recommended due to its ability to measure the importance of model inputs while accounting for their interactions. In this context, widely used global sensitivity analysis methods such as variance-based approaches (e.g., Sobol [68]) and screening-based methods (e.g., Morris [69]) can be employed.

4.3. Training and testing data preparation

4.3.1. Sampling influential model inputs space

Upon identifying influential and unobservable model inputs, the next step in the GenPhysiCal framework involves systematically sampling the model input space. The systematic sampling generates a range of input values covering several combinations of the identified unobservable model inputs. Suitable sampling methods should efficiently cover the sampling space with minimal samples, reducing the simulations needed to generate representative data. Ideally, the sampling procedure should balance comprehensive sample space coverage with fewer samples and capture the variability of the target variables in sufficient detail. Latin Hypercube Sampling (LHS) [70] is an effective method that meets these criteria. The method partitions the input space into non-overlapping intervals, uniformly samples values within each interval, and then combines these samples across dimensions to ensure uniform sampling of the entire input space [70,71]. Although LHS is the recommended choice in the GenPhysiCal framework, other sampling approaches with similar properties can also be used for sample generation.

4.3.2. Simulation and data processing

The simulation phase of the GenPhysiCal framework aims to produce simulated operational data by employing the physics-based model and the input samples generated in the framework's preceding phase. The process requires simulating using every combination of sampled model inputs while accounting for all other internal and external factors influencing the output variables of the physics model. This approach guarantees that, with each combination of sampled model inputs, the impact of these additional contributory factors is thoroughly considered

in generating representative data, which helps in accurately solving the calibration problem.

During the simulation phase of the GenPhysiCal framework, another crucial decision involves determining the simulation time resolution. Ideally, the time resolution should match that of the data collection by the sensors in the physical building, as the digital twin model needs to mirror the physical building's state as fast as the frequency of sensor observations. This decision should be tailored to the specific application of the digital twin and the primary function of the building.

Furthermore, the physics-based model employed for generating simulation data is presumed to reflect the physical building conditions with a satisfactory level of fidelity and has been validated. Therefore, the GenPhysiCal framework does not aim to address the intrinsic limitations of the physics model in precisely representing the characteristics of the physical building, as this accuracy is contingent on the assumptions made by the modelers.

The data generated from the execution of the forward model should then undergo preprocessing, such as data cleaning. Based on the characteristics of the generated simulation data, additional preprocessing methods may also be applied.

4.3.3. Data augmentation

Besides preprocessing, data augmentation is an integral component of the GenPhysiCal framework. The reliability of simulated data in accurately reflecting actual operational conditions in a physical building can be challenged by random or systematic errors within the highly uncertain building environment. These uncertainties are represented by the error ϵ in the calibration problem formulation. Therefore, despite the advantages of forward simulation in producing operational data for various unobservable model input combinations, it is essential to augment this data with additional samples that closely mirror real-world conditions. The augmented data helps ensure the generalizability of calibrator models to be trained using simulated data.

In the GenPhysiCal framework, the data augmentation stage is termed Virtual to Physical Observations Approximation (VPOA) within the context of a smart-building digital twin. It is essential to recognize that the augmentation phase does not aim to completely resolve the discrepancies between virtually simulated and physically measured observations. This task is exceptionally challenging due to the errors involved in physics model approximations of real-world conditions and diverse sources of uncertainty in the physical building. Addressing the modeling error, for example, falls within the purview of modelers. The primary focus of the GenPhysiCal framework is bridging the gap between the outputs of an existing validated physics model and the physical building measurements (i.e., continuous calibration of physics-based models), a gap primarily caused by the unobservability of influential and dynamic inputs.

The VPOA phase, therefore, only deals with augmenting the simulated observations with random noises of known distribution to mimic random errors in real-world building operations. The noisy observations in the actual building operation can be caused by systematic errors such as bias with constant deviations, drifting with time due to aging, the type and design of the sensors themselves, or random noises that vary due to the operational conditions of the sensors. The random variations can be mimicked by applying independently generated common random noises, such as Gaussian with predefined statistical properties, to the simulated data. This will allow the calibrator model to learn robust representations instead of solely relying on the simulated data pattern.

Another common condition in the physical building that can be imitated is the random failure of sensors, while the functional ones have some level of observation noise. Therefore, the second type of augmentation randomly masks the observable model output variables from the simulated data while applying random noise to the remaining observations, further ensuring a better approximation of the simulated data with the physical building conditions.

The final step in the VPOA phase involves merging the original simulated data with its augmented counterparts. Hence, there are three data set versions as per the proposed VPOA method: the original simulated data, simulated data with noisy observations, and a combination of random sensor failure and noise. The original simulated data is crucial for capturing the underlying input–output relationships as the physics-based model (forward model) dictates. In contrast, the augmented versions are instrumental in imitating the characteristics of observations from the physical building and improving the calibrator models' generalizability when deployed.

4.4. Calibrator model architecture and training

The GenPhysiCal framework employs generative inverse models to solve the calibration problem of physics-based models in smart-building digital twins. The use of generative models for solving the calibration problem is mainly due to their inherent ability to learn the underlying distribution from training data. Learning the underlying data distribution enables generative models to produce a full distribution of calibration solutions, thereby quantifying the associated uncertainty and making their predictions more reliable. In contrast, non-generative models such as Multi-Layer Perceptrons (MLP) learn a fixed mapping function from inputs to outputs, always producing the same point value predictions for the same input without capturing the underlying data distribution. As a result, generative models are more suitable for use as calibrator models.

Employing a generative inverse model for calibrating physics-based models differs from state-of-the-art approaches to accommodate the continuous calibration need in smart-building digital twins. State-of-the-art approaches typically use surrogate models that relate influential model inputs to the calibration target output variables to speed up the calibration process. In contrast, our approach, summarized in Fig. 3, directly addresses the calibration problem using a generative calibrator model. The calibrator model is systematically trained to learn the inverse mapping from available physical sensor measurements (y_o) collected from the building to the influential inputs (x^*) of the physics-based model ($M(x)$), which are not measured by physical sensors. By learning this inverse mapping from a finite number of evaluations of the physics-based model ($M(x)$), the generative calibrator model can estimate the full distribution of the influential inputs based on physical sensor measurements, eliminating the need for iterative methods such as genetic algorithms. This process is repeated with a defined time resolution, such as hourly, ensuring continuous model calibration that aligns the simulated outputs (y) of the physics-based model with the actual sensor measurements (y_o).

This study introduces a novel calibrator model named the Denoised-Encodings-Conditioned Invertible Neural Network (DECI-Net). The proposed model is based on a conditional invertible neural network combined with a denoising autoencoder conditioning network, as illustrated in Fig. 2. The architecture of DECI-Net is specifically designed to address three primary challenges in calibrating physics-based models: computational intensity and slowness, scalability limitations, and the quantification of uncertainty in calibration solutions, particularly under noisy real-world observations.

DECI-Net leverages the probabilistic generative capabilities of the cINN model for exact likelihood computation, bypassing the need for posterior distribution approximation. In addition, the bijective property of normalizing flows in invertible neural net architecture makes it particularly effective for inverse parameter estimation during calibration. DECI-Net is end-to-end trainable with simulated data, eliminating the need for costly likelihood model evaluations.

As depicted in Fig. 2, DECI-Net takes initial conditioning inputs y_o , representing the physically observable variables. The denoising autoencoder in DECI-Net refines the initial conditioning inputs y_o before propagating into the conditional coupling blocks of the model. This configuration stems from the need to handle noisy observations, such as

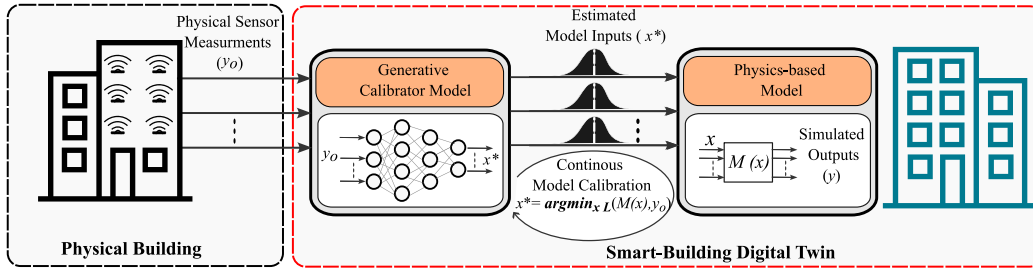


Fig. 3. Overall calibration process using a generative calibrator model.

sensor data with random variations and missing values. Therefore, instead of directly relying on the observable variables y_o for conditioning, DECI-Net compresses these inputs, learning relevant representations C_e in its bottleneck layer robust to noise and propagating such representation for conditioning, as shown in Fig. 2. Using latent representations C_e for conditioning in DECI-Net can also reduce the complexity of the cINN model, particularly in the case of high-dimensional conditioning inputs, which require stacking several coupling blocks. Such depth often leads to issues with vanishing or exploding gradients, as noted by Köhler et al. [72].

A combined loss function to train DECI-Net is proposed, as detailed in Eq. (8). This function comprises the log-likelihood loss of the cINN and a reconstruction loss for the denoising autoencoder, calculated as the mean squared error across the feature dimension of the autoencoder's input. In this loss formulation, x represents the unobserved model inputs estimated by the cINN, while z represents the target latent variable in the forward process. The hyperparameter λ in the loss function is a balancing factor. The reconstruction loss serves as a regularization term, enhancing the model's ability to maximize likelihood while simultaneously learning noise-robust representations from y_o . The combined loss term, even though it comes with an additional hyperparameter to be tuned, avoids a two-step training process of similar architectures [73]. In addition to the loss terms mentioned above, the L2 weight regularization term is included with the value controlled by a hyperparameter τ as suggested by Ardizzone et al. [56].

$$\mathcal{L}_{\text{DECI-Net}} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{2} \|f(x_i; C_{ei}, \theta)\|_2^2 - \log |J_i| \right) + \lambda \frac{1}{D} \|y_{oi} - \hat{y}_{oi}\|_2^2 + \tau \|\theta\|_2^2 \quad (8)$$

4.5. Calibrator model evaluation

Evaluating the generative calibrator model requires a comprehensive evaluation of its performance across various aspects. Such models function as probabilistic and point prediction models, depending on whether statistical summaries are applied to the predicted distributions. Point estimates and full distribution predictions are crucial in a smart-building digital twin physics-based model calibration context. Predicting the entire distribution is critical for quantifying uncertainty in unobservable model inputs estimated by the calibrator in the calibration process, providing the confidence level of predictions, which is vital for informed and reliable decision-making. On the other hand, when the trained calibrator model is to be used with deterministic physics-based models, validating the estimated calibration solutions requires the most probable point estimate (i.e., high-density point in the predicted distributions). Furthermore, as the primary purpose of the proposed generative model is for physics-based model calibration, measuring the agreement between the physics-based model outputs and the physical measurements using indicator distance metrics is needed. Finally, upon deployment in smart building digital twins, the calibrator models are expected to solve the calibration problem swiftly. The performance in this regard can be gauged by the inference time needed to obtain a single calibration solution.

In summary, a comprehensive evaluation of the generative calibrator model should focus on four aspects: its point estimate prediction accuracy, probabilistic prediction accuracy, model calibration accuracy (i.e., the agreement between the physics-model outputs and physically measured observations after re-evaluating using the estimated calibration solutions as inputs), and the time required for the calibrator model to solve for a single set of unknown model inputs during the calibration process (i.e., inference time).

4.6. Continuous physics-model calibration

The final step in the GenPhysiCal framework involves deploying the trained calibrator model to continuously calibrate the physics-based model, as detailed in Algorithm 1. The calibration process starts by collecting observable measurements y_o from the physical building and a set of n random samples from a standard normal distribution, denoted as z . The measurements, y_o , are then replicated n times to match with the z samples and get a full distribution prediction of size n . For each random sample from the latent distribution, the respective y_{oi} is fed into the encoder of DECI-Net. Subsequently, the inverse process of the DECI-Net, $f^{-1}(z_i)$, is employed to estimate the unobservable model inputs x_i , conditioned on the encoder's latent representation, which will be determined by the input to the encoder. During the inference stage of the DECI-Net model, the decoder is no longer needed, and only the encoder part of the denoising autoencoder will be used.

Once the predicted distribution of x is obtained, the maximum density point can be determined to identify the most probable point estimate (high-density point). This point estimate is then used to calculate the output y of the physics-based model $M(x)$ for validation. The final step involves computing the calibration error by comparing the computed y from the physics-based model with the measurements y_o to assess the model's accuracy in synchronizing the state of the physical building in its digital counterpart model. During a continuous physics-based model calibration process, a predefined error threshold $e_{\text{threshold}}$ is required to compare if the calibration error is under an acceptable range or not. In scenarios where the error is above the threshold, it indicates whether the calibrator or the physics-based model is not coping with changes in the physical building and requires a separate handling scheme, which is beyond the scope of this work.

5. Evaluation

The GenPhysiCal calibration framework was evaluated based on four performance criteria for calibrator models: point estimate prediction accuracy, probabilistic prediction accuracy, model calibration accuracy, and inference time. The subsequent subsections discuss the evaluation procedure.

5.1. Case study building

The proposed GenPhysiCal framework was evaluated using operational data generated from a building energy model due to the assumed unobservability of certain model inputs. In this study, the

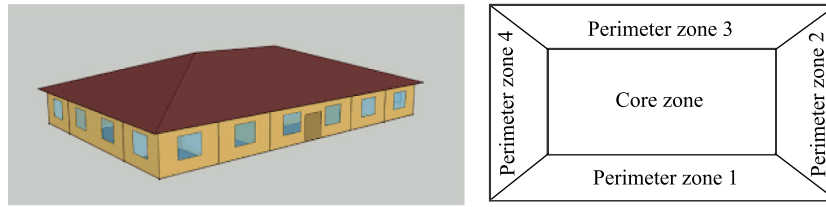


Fig. 4. 3D view of case study building (Left) and floor plan (Right).
Source: [75].

Algorithm 1: Continuous Physics-based Model Calibration Procedure.

Input: Trained DECI-Net Model, Physics-based model $y = M(x)$
Output: Calibration error between the model output y and physical measurements y_o

- 1 Obtain measurements y_o from the physical building;
- 2 Initialize z with n samples from the standard normal $\mathcal{N}(0, 1)$ distribution;
- 3 Duplicate the y_o values n times to match the count of z samples;
- 4 **for** $i = 1$ **to** n **do**
- 5 Forward pass y_{oi} into the encoder of DECI-Net;
- 6 Compute $f^{-1}(z_i)$ using the inverse process of the cINN part of DECI-Net;
- 7 Store the estimated value for unobservable model input x_i ;
- 8 **end**
- 9 Select the maximum density point from the estimated distribution of x ;
- 10 Compute the output y of the physics model $M(x)$ with the estimated x ;
- 11 Compute calibration error $e = L(y, y_o)$;
- 12 **if** $e < e_{\text{threshold}}$ **then**
- 13 Calibration Successful;
- 14 **end**
- 15 **else**
- 16 Calibration Unsuccessful;
- 17 **end**

open source building energy model provided by the US Department Of Energy (DOE) [74] was deployed for the case study as it enables future researchers to reproduce our work, and the reference building models are representative of small office buildings, a widely present building stock in North America. The specific reference building used for evaluation was assumed to be located in Atlanta, Georgia, USA, which has a single story with a total floor area of 511 m². The building has five zones, one core zone, and four perimeter zones. It uses gas for heating, electricity for cooling, plug loads, lighting, and a fan. The 3D model and floor plan of the case study building are depicted in Fig. 4.

5.2. Determining influential model inputs

Identifying influential model inputs within the GenPhysiCal framework is guided by the target physics-based model outputs that require calibration. Consequently, the targeted output variables for calibration in the case study building's energy model were the hourly electricity and gas consumption.

The case study building was assumed to lack sub-metering for lighting and miscellaneous plug loads, and it was also presumed to have no sensors for tracking occupant numbers in each zone. This selection of unobservable model inputs was based on occupants' significant yet uncertain impact and their behavior patterns on the building's energy consumption. In addition, occupant-dependent variables like equipment plug loads and lighting significantly affect the building's overall energy usage but are highly uncertain.

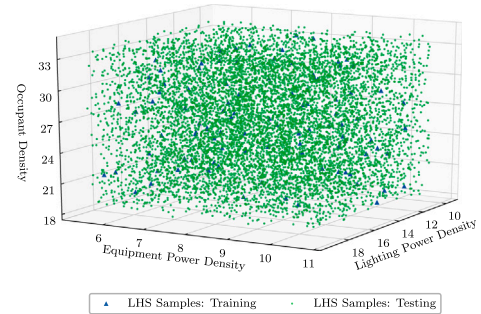


Fig. 5. 3D scatter plot showing LHS sampled inputs for training and testing data simulations.

In total, nine influential and unobserved model inputs were identified: three variables (occupant count, plug load, and lighting load) are common to both Zone-1 and Zone-3, another set is shared between Zone-2 and Zone-4, and three unique variables are specific to the Core zone.

5.3. Training and testing data preparation

5.3.1. Sampling influential inputs space

As previously identified, the three unobserved physics-model inputs were sampled using Latin Hypercube Sampling (LHS). The LHS samples for the unobserved model inputs were generated using area-dependent inputs in the building energy model, specifically occupant density (measured in m²/person), equipment power density (measured in W/m²), and lighting power density (measured in W/m²), these model inputs were varied from 50% to 100% of their default values.

The input LHS sample sizes were set at 400 for training and 8760 for testing and generated separately. The 8760 LHS samples for testing were to imitate the dynamic and uncertain conditions of the physical building by generating samples for every hour of the year. The uniform distribution of the sample area-dependent model inputs is shown in Fig. 5.

5.3.2. Simulation and data processing

The data generation process is shown in Fig. 6. The 8760 LHS samples were used with a single IDF file, actual weather data for 2022, and dynamic scheduled inputs to alter the model inputs during every hour of the simulation. Finally, the simulations for training and testing data produced 3,504,000 and 8760 data points, respectively.

The simulation data was subsequently processed to select physics model outputs that can be used for inverse estimation of the unobservable model inputs. Thus, fourteen output variables were identified as physically-observed model outputs. These include temperature and humidity measurements from five zones, an additional attic zone, and readings of building-level electricity and gas consumption.

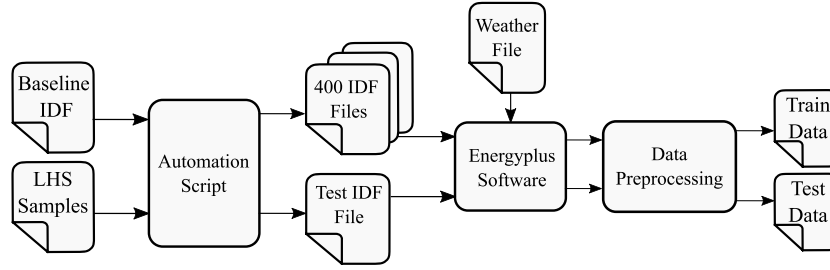


Fig. 6. Process of generating simulated training and testing data for evaluating calibrator models.

Table 1

Common configuration parameters and set values for cINN and DECI-Net calibrator models.

Configuration parameter	Set value
Hidden activation	ReLU
Output activation - $r(\cdot)$	Linear
Output activation - $S(\cdot)$	Tanh
Optimizer	Adam
Weight initializer	Xavier
Early stopping tolerance	5 Epochs
Number of epochs	100
Validation split	10% of train data

5.3.3. Data augmentation

Data augmentation was applied to the simulated training and testing datasets to imitate real-world uncertainties in building operations. These uncertainties are noisy measurements and missing readings resulting from sensor failures. As a result, two augmented datasets were created from the original training and testing data.

The first augmentation method added independent, standard Gaussian noise to the original training data, simulating noisy observations using a 10% noising factor. The second augmentation technique aimed to simulate sensor failures coupled with noise in the operational sensors. This involved randomly masking up to 35% (5 out of 14 variables) of the observations in each data row and adding noise to the unmasked readings. For both augmentation procedures, the noise was applied after standardizing the masked data combined with the original, ensuring that the noise scale remained proportional to the original dataset. The total data size tripled after the augmentation steps, resulting in data points three times the initial training and testing data.

5.4. Calibrator model training

In this study, standard Conditional Invertible Neural Network (cINN) and the proposed DECI-Net models were trained and tuned as calibrator models. The initial configuration parameters and values common to both models are shown in Table 1. A denoising autoencoder with three layers each in the encoder and decoder was employed for the DECI-Net model. Its hidden layers use ReLU activation, and the output layer uses linear activation.

The hyperparameters considered for tuning and their respective search spaces are summarized in Table 2. These search space values were chosen based on insights from multiple initial training.

5.5. Point estimate prediction accuracy metric

5.5.1. Root Mean Squared Error (RMSE)

The Root Mean Squared Error (RMSE) metric measures the standard deviation of residuals between the predicted and the actual ground truth values. RMSE is employed to evaluate the point estimate performance of the calibrator models in estimating the unobserved model inputs as solutions for the calibration problem. The mean of the predicted distributions was used as point estimates for each distribution

Table 2

Hyperparameter search spaces for cINN and DECI-Net.

Hyperparameter	Search space
Number of CCB	4, 8, 12, 16
Number of layers	2, 3
Number of neurons	64, 128, 256
L2 regularization	1e-2, 1e-3, 1e-4
DECI-Net - λ	1, 1e-1, 1e-2, 1e-3
Bottleneck size	4, 8, 10
Batch size	512, 1024, 2048
Learning rate	1e-2, 1e-3

targeting the ground truth value. The formula for RMSE is given in Eq. (9), where x_i represents the i th predicted value, $\hat{x}_i$ denotes the i th actual value, and n is the total number of data points [76].

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2} \quad (9)$$

5.6. Probabilistic prediction accuracy metrics

5.6.1. Probability calibration

The reliability of uncertainty estimates from the probabilistic calibrator models was evaluated by assessing the probability calibration accuracy of their predicted distributions against ground truth data using a calibration error metric [77]. Probability calibration refers to the statistical consistency of actual and predicted distributions [78,79] and is distinct from “model calibration accuracy”. Assessing probability calibration accuracy involved analyzing a cumulative distribution function (CDF), F_i , which targets the ground truth value y_i for each test point i , where N is the size of the test data.

A set of m confidence levels, $0 \leq p_1 < p_2 < \dots < p_m \leq 1$, was selected. For each level p_j , the empirical frequency $\hat{p}_j$ was computed as using Eq. (10) [77]. The number of test samples for which the predicted probability was less than or equal to p_j is then counted and averaged over the test dataset size.

The probability calibration performance can then be visualized using a calibration plot, which compares the threshold values p_j with the empirical frequencies $\hat{p}_j$ against a perfect calibration line represented by a diagonal line. Additionally, calibration performance is summarized by the calibration error (CE) metric, calculated as the sum of squared differences between p_j and $\hat{p}_j$, as shown in Eq. (11) [77].

$$\hat{p}_j = \frac{\sum_{i=1}^N |F_i(y_i) \leq p_j|}{N} \quad (10)$$

$$\text{CE} = \sum_{j=1}^m (p_j - \hat{p}_j)^2 \quad (11)$$

5.6.2. Sharpness of predicted distributions

The sharpness of probabilistic forecasts was assessed using the mean of the central prediction intervals at different nominal coverage rates $(1 - \alpha)$ given by Eq. (12), where quantile $(\hat{q}_{\alpha/2})$ denotes the upper

bound, and $\hat{q}_{1-\alpha/2} - \hat{q}_{\alpha/2}$ denotes the lower bound of the prediction interval [80,81].

$$S^\alpha = \frac{1}{N} \sum_{i=1}^N (\hat{q}_{1-\alpha/2} - \hat{q}_{\alpha/2}) \quad (12)$$

5.6.3. Continuous Ranked Probability Score (CRPS)

In evaluating probabilistic model performance, the Continuous Ranked Probability Score (CRPS) is recognized as a strictly proper metric. CRPS, a generalization of the mean absolute error for probabilistic models, was employed for comparing the baseline and DECI-Net models [79]. As a summary skill score, CRPS evaluates the predictive performance of probabilistic models, addressing both probability calibration and sharpness [78]. It measures the discrepancy between the predictive cumulative distribution function $\hat{F}_{pred}^i$ and the actual ground truth cumulative distribution function $F_{x_{act}}^i$, as illustrated in Eq. (13) [79,81]. The ground truth CDF, $F_{x_{act}}^i$, is represented as a step function, being one when $x \geq x_{act}$ and zero otherwise [79]. The CRPS values, computed for each sample i , are then averaged over N samples.

$$CRPS = \frac{1}{N} \sum_{i=1}^N \int_{-\infty}^{+\infty} \left(\hat{F}_{pred}^i(x) - F_{x_{act}}^i(x) \right)^2 dx \quad (13)$$

5.7. Model calibration accuracy metric

5.7.1. Coefficient of Variation of Root Mean Squared Error (CVRMSE)

The Coefficient of Variation of the Root Mean Square Error (CVRMSE) is a standard metric for evaluating model calibration accuracy in building energy models [27]. CVRMSE was used to evaluate the model calibration accuracy, which measures how closely the simulated outputs of the physics-based model after being re-evaluated using the estimated calibration solution as input, align with measured data. The CVRMSE metric was calculated as per Eq. (14) [26,27], where s_i and m_i represent the simulated and measured data respectively, n is the number of data points, $\bar{m}$ is the mean of the measured data, and p is suggested to be one [82].

$$CVRMSE = \frac{1}{\bar{m}} \sqrt{\frac{\sum_{i=1}^n (m_i - s_i)^2}{n - p}} \times 100 (\%) \quad (14)$$

According to the guidelines from the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) and the Federal Energy Management Program (FEMP), the calibration accuracy threshold for building energy models for the hourly CVRMSE is set at 30% [83,84]. This study's target in terms of model calibration performance was to achieve error values well below the hourly CVRMSE value set by ASHRAE and FEMP.

5.8. Experiments

Three experiments were conducted to evaluate the baseline cINN and DECI-Net calibrator models. The goal was to rigorously assess the calibrator models' ability to estimate unobserved influential inputs, ensuring precise, fast, and ongoing synchronization between the physical building and its digital twin while accounting for buildings' uncertain environments. In each of the three experiments, the point estimate prediction accuracy, probabilistic prediction accuracy, model calibration accuracy, and inference time were measured.

The three experiments designed to assess the performance of the calibrator models are summarized in Table 3. In the first experiment (Experiment 1), the baseline cINN model and DECI-Net were compared using unmodified simulated data; no missing sensor or noisy observations were included on the physically observable variables. This experiment evaluated the calibrator models' capability to capture underlying relationships among variables from the simulated dataset.

The second experiment (Experiment 2) was designed to evaluate the calibrator models in conditions where all sensors measuring physically observed variables are functional but subjected to noise. Finally, the third experiment (Experiment 3) combines random sensor malfunctions and noise in the remaining functional sensors.

Table 3

Summary of experiments and the types of test data considered. The symbols ✓ and * denote inclusion and exclusion, respectively.

Experiment	Unmodified test data	Noised test data	Missing test data
Experiment 1	✓	*	*
Experiment 2	*	✓	*
Experiment 3	*	✓	✓

5.9. Results and discussion

The results discussed in subsequent subsections are from the best-performing calibrator models after hyperparameter tuning. Specifically, the hyperparameters for the best-performing DECI-Net model included 12 conditional coupling blocks, two layers in each coupling block, 64 neurons per layer, L2 regularization of $1e-4$, a batch size of 2048, a learning rate of $1e-3$, and a denoiser bottleneck size of 8.

The tuned hyperparameter combinations for the standard cINN model included 16 conditional coupling blocks, two layers in each block with 64 neurons per layer, L2 regularization of $1e-4$, a batch size of 2048, and a learning rate of $1e-3$.

5.9.1. Point estimate prediction accuracy

The probabilistic predictions from the standard cINN and DECI-Net calibrator models were approximated using the mean of the entire distribution to derive a point estimate. Subsequently, RMSE values were calculated for each zone and each unobserved model input (Occupant Count, Lighting Load, Plug Load) across all three experiments.

The summarized results in Table 4 reveal three key observations about the point estimate performance of the calibrator models. Firstly, in all three experiments, for every unobserved input across all building zones, the DECI-Net model's RMSE score consistently outperformed that of the standard cINN model, as highlighted in the table. Secondly, in the first experiment with neither noise nor missing sensor readings, both cINN and DECI-Net demonstrated similar performance levels, though DECI-Net achieved better RMSE values. In the second and third experiments, under noisy observations and the combined presence of missing sensors and noise, the standard cINN model's accuracy declined significantly. In contrast, the DECI-Net model only slightly decreased its prediction accuracy.

In addition to the RMSE, density plots of predicted mean values by both calibrator models are shown in Figs. 7, 8, and 9. Both calibrator models generally were able to capture the overall shape of the actual distribution of unobservable model inputs, though noticeable overestimations in the density plots of occupant count (in the 4–6 range) and plug load (in the 0.2–0.6 range) were observed in the first two experiments. In all cases, DECI-Net's overestimated peaks were consistently lower than those of the standard cINN model.

The density plots of Experiment 3 shown in Fig. 9 showed highly skewed distribution in the standard cINN model's predictions, falling beyond the actual distribution ranges. Furthermore, cINN's overestimated peaks significantly diverged from the actual peaks. Conversely, the DECI-Net demonstrated stable performance under noise and missing sensor data, as shown in Fig. 9.

These results indicate that the standard cINN model struggled with the uncertain sensing conditions of the building. In contrast, the DECI-Net maintained consistent performance due to its denoising autoencoder conditioning network. This network effectively mapped noisy data into latent conditions that were more resistant to noise and missing data.

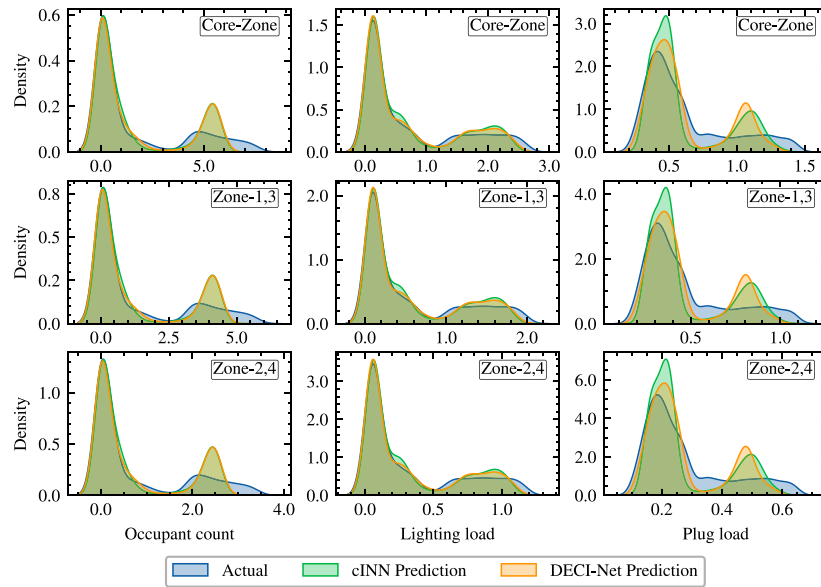
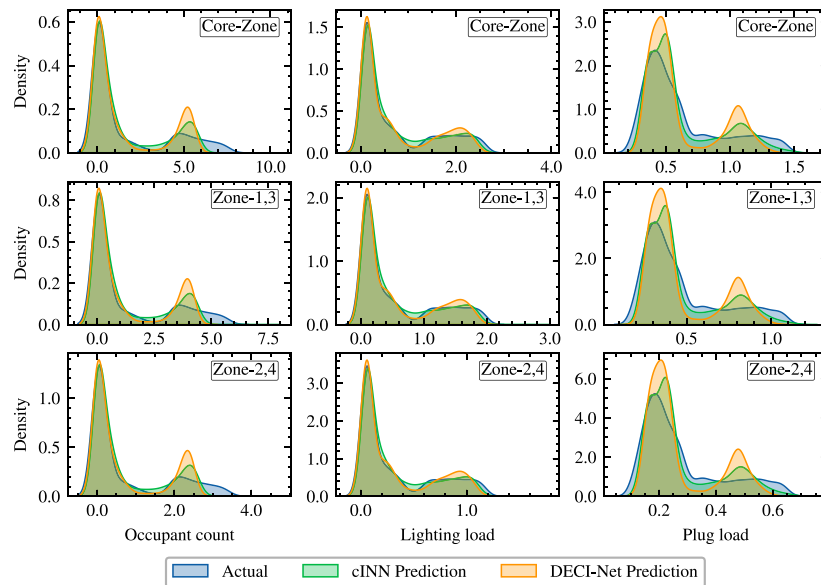
5.9.2. Probabilistic prediction accuracy-probability calibration

The reliability of uncertainty estimates for unobservable inputs of the building energy model by both the standard cINN and DECI-Net models is summarized in the calibration plots depicted in Figs. 10, 11, and 12.

Table 4

RMSE Values for Mean Predictions of Occupant Count (People), Lighting, and Plug Load (kW) by cINN and DECI-Net Calibrator Models for All Building Zones.

Building Zones	Model Inputs	Experiment-1		Experiment-2		Experiment-3	
		cINN	DECI-Net	cINN	DECI-Net	cINN	DECI-Net
Core-Zone	Occupant Count	0.78	0.74	1.437	0.894	3.003	0.95
	Lighting Load	0.158	0.151	0.386	0.198	1.017	0.212
	Plug Load	0.122	0.116	0.2	0.135	0.422	0.139
Zone-1,3	Occupant Count	0.591	0.561	1.089	0.678	2.277	0.721
	Lighting Load	0.12	0.114	0.293	0.15	0.771	0.161
	Plug Load	0.092	0.088	0.151	0.103	0.32	0.105
Zone-2,4	Occupant Count	0.351	0.333	0.646	0.402	1.35	0.427
	Lighting Load	0.071	0.068	0.174	0.089	0.458	0.094
	Plug Load	0.055	0.052	0.09	0.061	0.19	0.063

**Fig. 7.** Density plots of the predicted mean values for occupant count (People), lighting, and plug load (kW) by cINN and DECI-Net models in conditions without noise or missing sensor measurements (Experiment 1).**Fig. 8.** Density plots of the predicted mean values for occupant count (People), lighting, and plug load (kW) by cINN and DECI-Net models in conditions with noisy sensor measurements (Experiment 2).

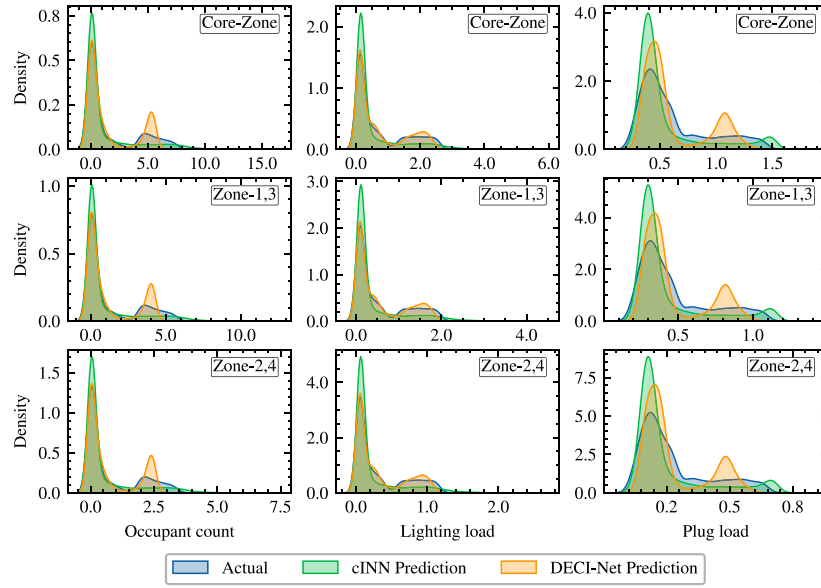


Fig. 9. Density plots of the predicted mean values for occupant count (People), lighting, and plug load (kW) by cINN and DECI-Net models in conditions with noisy and missing sensor measurements (Experiment 3).

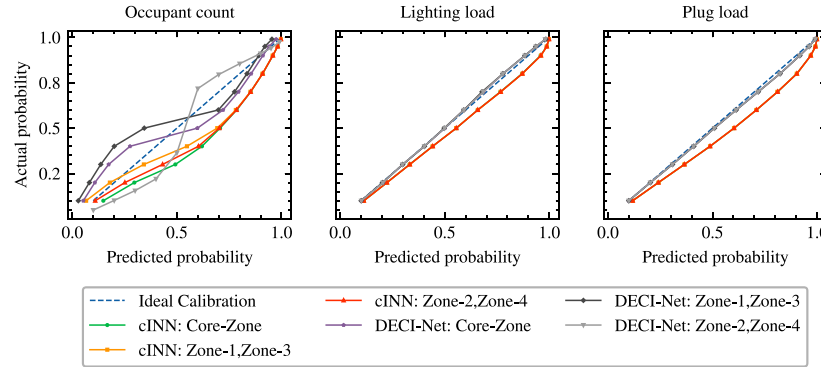


Fig. 10. Calibration plot comparing cINN and DECI-Net calibrator models in conditions without noise or missing sensor measurements (Experiment 1).

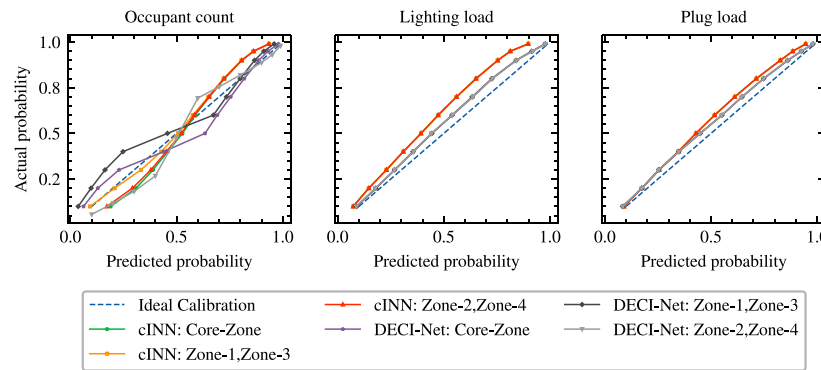


Fig. 11. Calibration plot comparing cINN and DECI-Net calibrator models in conditions with noisy sensor measurements (Experiment 2).

The calibration plots in all three experiments of the occupant count showed the predicted probabilities below and above the actual probabilities. In Experiment 1, DECI-Net showed overconfidence, and deviations from the ideal calibration line showed underconfidence. However, the DECI-Net model was still able to match the actual probabilities at 0.9, 0.95, and 0.99, while cINN showed underconfidence in a broad range of 0.4 to 0.99.

The calibration plots of occupant count for Experiments 2 and 3 also demonstrated that DECI-Net still closely matched the actual probabilities in the 0.9–0.99 range. The cINN model exhibited comparable performance for Experiment 2. However, for Experiment 3, it was way far above the ideal calibration line, proving unreliable.

In Experiment 1, for the remaining unobserved model inputs – Lighting Load and Plug Load – the DECI-Net model's predicted probabilities

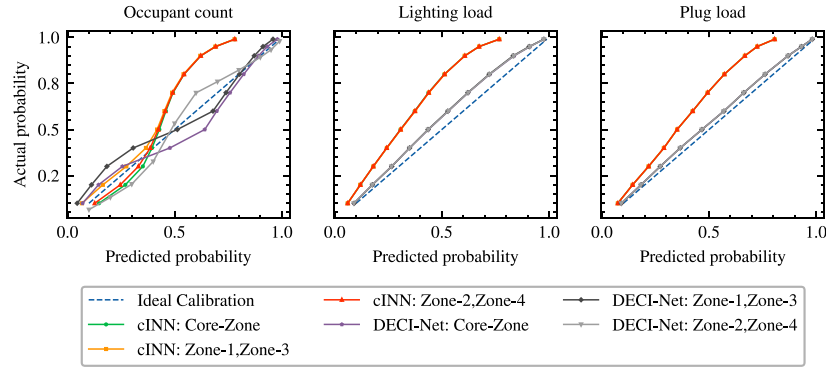


Fig. 12. Calibration plot comparing cINN and DECI-Net calibrator models in conditions with noisy and missing sensor measurements (Experiment 3).

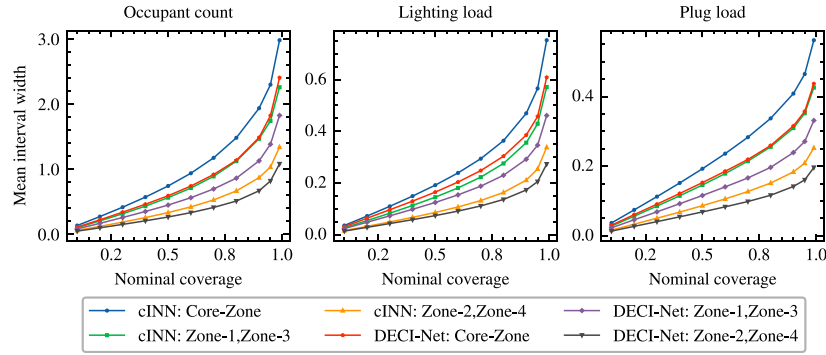


Fig. 13. Sharpness plot comparing cINN and DECI-Net calibrator models in conditions without noisy or missing sensor measurements (Experiment 1).

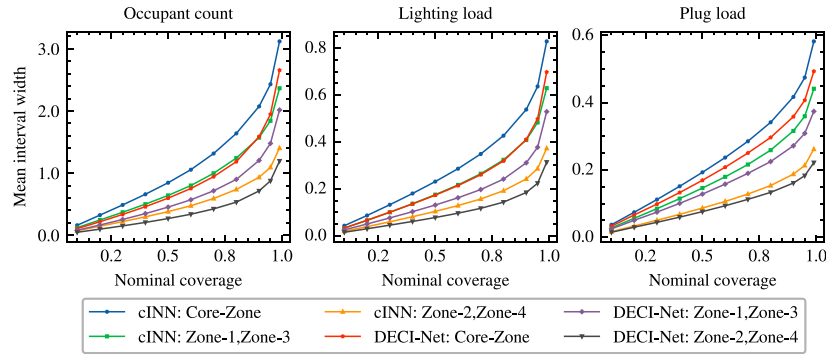


Fig. 14. Sharpness plot comparing cINN and DECI-Net calibrator models in conditions with noisy sensor measurements (Experiment 2).

were perfectly calibrated, showcasing highly reliable uncertainty quantification in its estimated solutions for calibrating the building energy model. Contrarily, the standard cINN model showed underconfidence in its predictions for these inputs. In Experiment 2, both models exhibited overconfidence, but DECI-Net aligned better with the ideal calibration line. During Experiment 3, the standard cINN model demonstrated marked overconfidence, significantly deviating from the ideal line, while the DECI-Net model showed notably better adherence to the ideal calibration line.

Summarizing key findings from the three experiments, the DECI-Net calibrator model demonstrated superiority in accurately predicting unobservable inputs of the building energy model. It effectively quantified uncertainties in its estimated solutions, even in the presence of noisy measurements used for inverse estimation. Under conditions of missing data and noise, DECI-Net closely matched actual probabilities, outperforming the standard cINN model, while its accuracy could be further improved with tuning. Overall, these results highlight the effectiveness and potential of DECI-Net as a calibrator model for physics-based models in smart building digital twins.

5.9.3. Probabilistic prediction accuracy-sharpness of predicted distributions

Besides assessing the calibration plots of probabilistic predictions by both calibrator models, the sharpness of the predicted distributions was also evaluated. The sharpness plots are illustrated in Figs. 13, 14, and 15. These results further underscore the effectiveness of the DECI-Net model. Across all three experiments, DECI-Net demonstrated narrower mean interval widths in its predicted distributions compared to the standard cINN model, thereby reinforcing the robustness of the proposed model.

5.9.4. Probabilistic prediction accuracy-CRPS

Lastly, the overall probabilistic prediction performance of the two competing models (the standard cINN and DECI-Net) in estimating the unobserved building energy model inputs was evaluated using a strictly proper evaluation metric, CRPS, which summarizes overall performance. The results for CRPS values are presented in Table 5. In all three experiments, in every zone, and for all unobservable model inputs, the DECI-Net model consistently outperformed the standard cINN model.

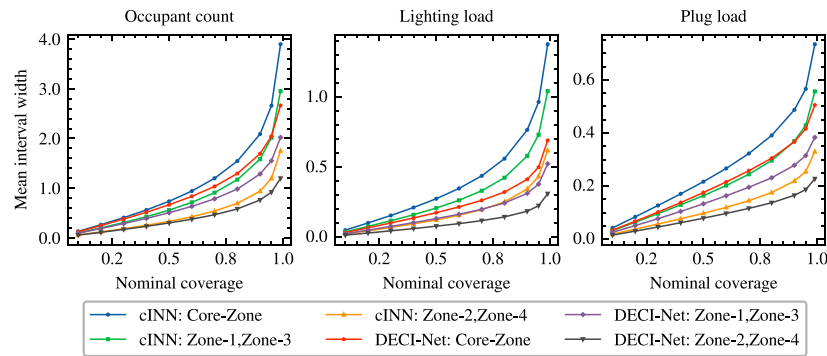


Fig. 15. Sharpness plot comparing cINN and DECI-Net calibrator models in conditions with noise and missing sensor measurements (Experiment 3).

Table 5

Comparison of CRPS scores between cINN and DECI-Net calibrator models.

Building Zones	Model Inputs	Experiment-1		Experiment-2		Experiment-3	
		cINN	DECI-Net	cINN	DECI-Net	cINN	DECI-Net
Core-Zone	Occupant Count	0.302	0.279	0.584	0.333	1.615	0.355
	Lighting Load	0.072	0.067	0.171	0.086	0.564	0.091
	Plug Load	0.062	0.056	0.102	0.071	0.239	0.072
Zone-1,3	Occupant Count	0.229	0.211	0.443	0.253	1.224	0.269
	Lighting Load	0.054	0.051	0.13	0.065	0.428	0.069
	Plug Load	0.047	0.042	0.077	0.054	0.181	0.055
Zone-2,4	Occupant Count	0.136	0.125	0.263	0.15	0.726	0.159
	Lighting Load	0.032	0.03	0.077	0.039	0.254	0.041
	Plug Load	0.028	0.025	0.046	0.032	0.107	0.033

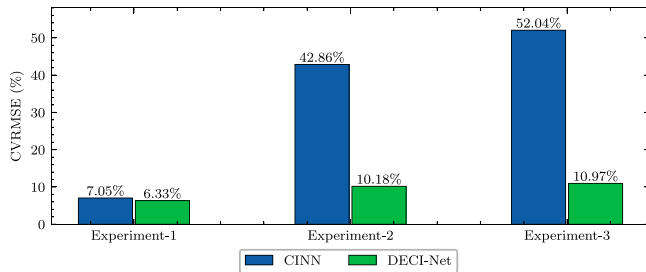


Fig. 16. CVMSE comparison of cINN and DECI-Net calibrator models in calibrating electricity consumption for the case study building.

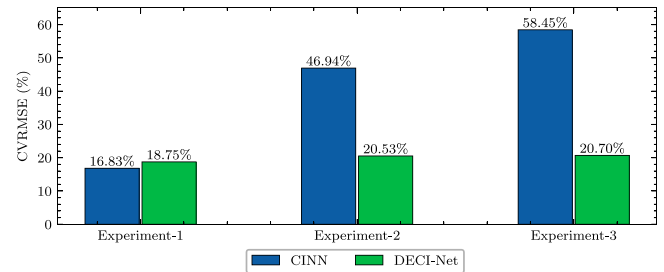


Fig. 17. CVMSE comparison of cINN and DECI-Net calibrator models in calibrating gas consumption for the case study building.

5.10. Model calibration accuracy

The model calibration accuracy was evaluated to measure how closely the building energy model outputs matched the measured data when using the estimated unobservable model inputs. The mean predictions of the two calibrator models in all three experiments were input into the building energy model. The resulting electricity and gas consumption outputs of these simulations were then compared with the initial simulated outputs. The CVMSE scores of both calibrator models are shown in Figs. 16 and 17 for hourly electricity and gas consumptions respectively.

In calibrating the electricity consumption output of the building energy model, DECI-Net demonstrated better accuracy than the standard cINN calibrator model, as reflected by their hourly CVMSE values. Both calibrator models successfully exceeded the standard hourly threshold established by FEMP and ASHRAE, achieving CVMSE values of 7.05% and 6.33%, respectively. Regarding model calibration accuracy in Experiments 2 and 3, DECI-Net kept its CVMSE values under the 30% threshold, slightly increasing to 10.18% in Experiment 2 and 10.97% in Experiment 3. On the other hand, the standard cINN calibrator model significantly underperformed, exceeding the standard

threshold by more than 12% and 22%, respectively, indicating that its performance in calibrating the physics model falls below the acceptable standard.

In calibrating the facility's gas consumption, both models exhibited higher hourly CVMSE values compared to their performance in electricity consumption evaluations. Specifically, in Experiment 1, both models achieved an hourly CVMSE under 30%, meeting the set requirement. In the subsequent two experiments, the calibrated building energy model's accuracy with DECI-Net predictions remained below the 30% threshold, satisfying the standard requirements. However, the standard cINN model exceeded this threshold by more than 16% and 28% in these experiments.

Overall, the DECI-Net calibrator model demonstrated superior performance in calibrating building energy model outputs compared to the standard cINN model while meeting accepted standards for calibration accuracy. The results effectively validate the proposed DECI-Net model's ability to calibrate physics-based models and provide precise uncertainty estimates in its calibration solutions, both instrumental in smart-building digital twins.

5.11. Inference time

The average inference time for computing the calibration solution per observation by the calibrator models (cINN and DECI-Net) is 0.038 second for the standard cINN model and 0.043 second for the DECI-Net model. The time measurements did not vary across experiments for each of the experimental cases (i.e., Experiment 1 through Experiment 3). The inference time measurements demonstrate that once trained and fine-tuned, the DECI-Net calibrator model can efficiently solve the calibration problem without resource-intensive searches in a very short amount of time.

5.12. Limitations of GenPhysiCal framework

The GenPhysical framework has two main limitations. The first limitation is its limitation in dealing with unsuccessful calibration attempts by the calibrator model once deployed. Such scenarios can result from various factors in physical buildings that the model did not encounter during the training process. This study used data augmentation procedures that mimic random errors and sensor failures in the real world to enhance the generalizability of the calibrator model to these unseen scenarios. However, given the complex dynamics of real-world buildings, unforeseen situations such as physical changes to the building not considered during training may occur. Accurately replicating all possible scenarios in the physical building is extremely challenging. Therefore, an adaptive ongoing learning mechanism should be in place so that cases where the calibrator model fails to satisfy the minimum calibration error threshold can be used to fine-tune its performance.

The second limitation is related to the nature of conditional invertible neural networks (cINN). Although cINN models are very powerful, they have expressiveness limitations when learning very complex data distributions. This limitation arises mainly due to the need to maintain invertibility, which restricts possible transformations. Consequently, more expressive cINN models may need to be deep, involving several stacked coupling blocks. In such cases, sophisticated training schemes may be required to address challenges like vanishing or exploding gradients.

5.13. Practical implementation challenges

The practical implementation of the GenPhysiCal framework requires preparation in both physics-based modeling and the sensing infrastructure of the physical building. The physics-based model should have sufficient fidelity to accurately relate physically measured variables to other building properties. Additionally, the model must be validated to ensure it accurately represents the actual building operation. Model validation is especially important when transforming physics-based models from the design to the operational phase.

Another requirement is sufficient observability of the physical building through physical sensors. However, due to the physical sensing limitations detailed in this study, observability can be significantly diminished during operation if multiple sensors fail or provide erroneous observations simultaneously. Although the calibrator model is designed to function under limited observability and with random errors in observations, a higher number of simultaneous sensor failures and errors can significantly challenge the practical implementation of the GenPhysiCal framework.

6. Conclusion

This study introduced a comprehensive framework named GenPhysiCal, tailored to the continuous model calibration needs of smart-building digital twins. The GenPhysiCal framework included problem formulation, systematic data generation, training, evaluation, and continuous model calibration procedures. A novel generative calibrator

model architecture called DECI-Net was presented to enable a continuous, fast, and scalable approach for calibrating multiple model outputs with measured observations while quantifying uncertainty in the estimated calibration solutions. In addition, DECI-Net was specifically devised to deal with the uncertain sensing environment of buildings, characterized by random sensor failures and measurement noise.

This study also introduced a continuous model calibration procedure that utilizes ongoing sensor data collection to estimate dynamic and unobservable model inputs. The continuous calibration procedure ensures the alignment of multiple model outputs with measured observations using the pre-trained generative calibrator model, avoiding computationally expensive model evaluations during the calibration process.

Comprehensive evaluation based on various experiments confirmed the effectiveness of the GenPhysiCal framework in addressing the challenges of continuous model calibration. The experimental results showed that the generative calibrator model (DECI-Net) provided fast, simultaneous, and reliable calibration solutions under randomly malfunctioning sensors and noisy observations, with a slight decrease in performance. In our experiments, the calibrator model estimated nine unknown model inputs in an average time of 0.043 second to calibrate the electricity and gas consumption of the case study building. As measured using the Coefficient of Variation Root Mean Squared Error (CVRMSE) score, the calibration accuracies achieved are 6.33%, 10.18%, and 10.97% for electricity, and 18.75%, 20.53%, and 20.7% for facility-level gas consumption without missing sensors or noisy observations, with noisy measurements, and with missing sensors and noisy observations, respectively.

This work will continue to explore methods for detecting changes in the physical building not initially accounted for in the original physics-based model and procedures for updating the calibrator model with new data through online training schemes. Additionally, the transferability of calibrator models from a source building to a target building's digital twin for minimizing training efforts will be investigated.

CRediT authorship contribution statement

Dagimawi D. Eneyew: Conceptualization, Methodology, Investigation, Formal analysis, Visualization, Software, Writing – original draft. **Miriam A.M. Capretz:** Conceptualization, Methodology, Funding acquisition, Resources, Writing – review & editing.. **Girma T. Bitsuamlak:** Conceptualization, Methodology, Funding acquisition, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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